

ALGORITHMS & COMPLEXITY

LECTURE 6

For $\boxed{k=3}$ $\boxed{\dim(V_\infty^3) = 3}$

We may have in:

$$\boxed{a_n = d_1 a_{n-1} + d_2 a_{n-2} + d_3 a_{n-3}} \quad \Delta$$

with $r = C r^n$

$$\boxed{r^3 - d_1 r^2 - d_2 r - d_3 = 0} \quad (0)$$

characteristic equation corresponding to scheme Δ

Case A 3 real roots ^{to (0)} different:
 $r_1 \neq r_2 \neq r_3$ (multiplicity of each root is 1)

$$\Rightarrow \boxed{a_n = C_1 r_1^n + C_2 r_2^n + C_3 r_3^n} \quad \begin{array}{l} \text{basis of } V_\infty^3 \\ \text{depends on parameters} \end{array}$$

once $\hat{a}_0, \hat{a}_1, \hat{a}_2$ are given we eliminate 3 parameters from *

$$* \begin{cases} \hat{a}_0 = a_0 = C_1 r_1^0 + C_2 r_2^0 + C_3 r_3^0 \\ \hat{a}_1 = a_1 = C_1 r_1^1 + C_2 r_2^1 + C_3 r_3^1 \\ \hat{a}_2 = a_2 = C_1 r_1^2 + C_2 r_2^2 + C_3 r_3^2 \end{cases}$$

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& after computing from * $\tilde{C}_1, \tilde{C}_2, \tilde{C}_3$
the unique solution

$$a_n = \tilde{C}_1 r_1^n + \tilde{C}_2 r_2^n + \tilde{C}_3 r_3^n$$

CASE B $r_1 = r_2 = r$ & r_3 ^{← multiplicity 1} different from r
multiplicity 2 ↑ roots to (•)

$$a_n = C_1 r^n + C_2 n r^n + C_3 r_3^n$$

if $\hat{a}_0, \hat{a}_1, \hat{a}_2$ given we can eliminate parameters C_1, C_2 & C_3 from

$$\begin{cases} \hat{a}_0 = a_0 = C_1 r^0 + C_2 \cdot 0 r^0 + C_3 r_3^0 \\ \hat{a}_1 = a_1 = C_1 r^1 + C_2 \cdot 1 \cdot r^1 + C_3 r_3^1 \\ \hat{a}_2 = a_2 = C_1 r^2 + C_2 \cdot 2 \cdot r^2 + C_3 r_3^2 \end{cases}$$

once $\tilde{C}_1, \tilde{C}_2$ & $\tilde{C}_3$ are computed
a unique solution to $\hat{a}_0, \hat{a}_1, \hat{a}_2$
given reads as

$$a_n = \tilde{C}_1 r^n + \tilde{C}_2 n r^n + \tilde{C}_3 r_3^n$$

Case C $r_1 = r_2 = r_3 = r$ (multiplicity of
roots to (•) 3)

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Then

$$a_n = C_1 r^n + C_2 n r^n + C_3 n^2 r^n$$

Solution depends on 3 parameters ($\dim V_\omega^3 = 3$). If $\hat{a}_0, \hat{a}_1$ & $\hat{a}_2$ are given then we can eliminate C_1, C_2, C_3 from:

$$\begin{cases} \hat{a}_0 = a_0 = C_1 r^0 + C_2 \cdot 0 r^0 + C_3 0^2 r^0 \\ \hat{a}_1 = a_1 = C_1 r^1 + C_2 1 \cdot r^1 + C_3 1^2 r^1 \\ \hat{a}_2 = a_2 = C_1 r^2 + C_2 2 \cdot r^2 + C_3 2^2 r^2 \end{cases}$$

Again upon eliminating & computing $\tilde{C}_1, \tilde{C}_2, \tilde{C}_3$ we obtain a unique solution

$$a_n = \tilde{C}_1 r^n + \tilde{C}_2 n r^n + \tilde{C}_3 n^2 r^n$$

Example ^(a) $a_n = 2a_{n-1} + 5a_{n-2} - 6a_{n-3}$

once we substitute $a_n = C r^n$ we obtain a characteristic polynomial for

$$r^3 - 2r^2 - 5r + 6 = (r-1)(r+2)(r-3) = 0$$

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ATTENTION finding roots of schemes of order > 2 generates a technical difficulty in finding roots of the corresponding characteristic polynomial. The higher order k is the higher order of polynomial we have.

Hints if we search for roots of polynomial of integer coefficients & the roots are to be integer they must be divisors of a_0 where

$$W_n(x) = a_n x^n + a_{n-1} x^{n-1} + \dots + a_1 x + a_0.$$

Once you find one root $\bar{x}$ we divide $W_n(x)$ by $(x - \bar{x})$ and get a polynomial of degree $n-1$ i.e. $W_n(x) = (x - \bar{x}) \cdot \hat{W}_{n-1}(x)$ & we continue the procedure.

So
$$a_n = C_1 1^n + C_2 (-2)^n + C_3 3^n \quad *$$

if initial conditions are not given then a_n is of the form (*) & depends on 3 parameters.

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$$b) \begin{cases} b_{n+1} = 5b_n - 3b_{n-1} - 9b_{n-2} \quad (*) \\ b_0 = 6 \quad b_1 = -8 \quad b_2 = -22 \quad (**) \end{cases}$$

Characteristic equation

$$\boxed{r^3 - 5r^2 + 3r + 9 = 0}$$

$$(r+1)(r-3)^2 = 0$$

$$\Rightarrow \boxed{b_n = C_1(-1)^n + C_2 3^n + C_3 n 3^n}$$

Taking into account (**) $\uparrow$ general solution to *

$$\begin{cases} C_1 + C_2 = 6 \\ -C_1 + 3C_2 + 3C_3 = -8 \\ C_1 + 9C_2 + 18C_3 = -22 \end{cases}$$

$$\Rightarrow C_1 = 5, \quad C_2 = 1, \quad C_3 = -2$$

$$\text{hence: } \boxed{b_n = 5(-1)^n + 3^n - 2n 3^n}$$

In general:

$$(*) \quad a_n = \sum_{i=1}^k c_i a_{n-i}$$

for $k=4$

$$r_1 = r_2 = r_3 = r_4$$

$$r_1 = r_2 \neq r_3 = r_4$$

$$r_1 = r_2 = r_3 \neq r_4$$

$$r_1 \neq r_2 \neq r_3 \neq r_4$$

$\Rightarrow$ characteristic polynomial

$$W_k(r) = r^k - \sum_{i=1}^k c_i r^{k-i}$$

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Th

Assume characteristic polynomial $W_k(r)$ has $r_1, r_2, \dots, r_{l_k}$ different roots (real or complex) each with the multiplicity $m_1, m_2, \dots, m_{l_k}$ such that $m_1 + m_2 + \dots + m_{l_k} = k$ so

$$W_k(r) = \prod_{i=1}^{l_k} (r - r_i)^{m_i}$$

Then each solution to (*) is a linear combination of

$$\begin{aligned} & r_1^m, m r_1^{m-1}, \dots, m^{m_1-1} r_1^m \\ & r_2^m, m r_2^{m-1}, \dots, m^{m_2-1} r_2^m \\ & \vdots \\ & r_{l_k}^m, m r_{l_k}^{m-1}, \dots, m^{m_{l_k}-1} r_{l_k}^m \end{aligned}$$

which depends on $c_1, c_2, \dots, c_k$ parameters. If additionally $\hat{a}_0, \hat{a}_1, \dots, \hat{a}_{k-1}$ are given then $c_1, c_2, \dots, c_k$ can be computed from the respective system of k -eqns in k -unknowns.

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Still we do not know how to solve inhomogeneous difference scheme — in particular we do not know how to handle Hanoi Tower problem:

$$T_n = 2T_{n-1} + 1$$

In general how to deal with

$$\bullet_{IH} \left[a_n = \sum_{i=1}^K c_i a_{n-i} + \underbrace{f(n)}_{\neq 0} \right] \text{ inhomogeneous linear scheme with constant coefficients.}$$

We know from $a_n = f(a_{n-1}, \dots, a_{n-k}, n)$ (1)
if $a_0, a_1, \dots, a_{k-1}$ are given then has unique solution. So a_n depends on k -parameters.

$$\bullet_H \quad h_n = \sum_{i=1}^K c_i h_{n-i} \quad \begin{array}{l} \text{the corresponding} \\ \text{homogeneous scheme} \\ h_n \leftarrow \text{depends on } k\text{-parameters} \end{array}$$

b_n is a special solution to $\bullet_{IH}$ then

$a_n = h_n + b_n$ satisfies $\bullet_{IH}$ & depends on k -parameter

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& vice versa if a_n ^{depends on k-param.} satisfies $\bullet_{IH}$

we take b_n a special solution

$$a_n = \underbrace{a_n - b_n}_{\text{satisfies } \bullet_H} + b_n \leftarrow \text{a special sol. of } \bullet_{IH}.$$

satisfies $\bullet_H$ (depends on k-param.)

Th Assume:

b_n is a specific solution to $\bullet_{IH}$.

Then a_n is a general solution of $\bullet_{IH}$

$$\Leftrightarrow b_n = a_n + h_n$$

where h_n is the general solution of $\bullet_H$.

So

$$(1) \left[\begin{array}{l} a_n = \sum_{i=1}^k c_i a_{n-i} + f(n) \\ a_0, a_1, \dots, a_{k-1} \text{ [OPTIONALLY GIVEN]} \end{array} \right.$$

SOLUTION FOR (1)

$$I \left[\begin{array}{l} h_n = \sum_{i=1}^k c_i h_{n-i} \\ \text{Note: } h_n \text{ depends on } k \text{ parameters} \end{array} \right. \quad \begin{array}{l} \text{solve first} \\ \text{homogeneous} \\ \text{system} \end{array}$$

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II Find a specific solution for inhomogeneous equation

$$b_n = \sum_{i=1}^k c_i a_{n-i} + f(n)$$

 prediction method $\leftarrow$ see later

III general solution to inhomogeneous eqn

$$a_n = \underbrace{h_n}_I + \underbrace{b_n}_{II}$$

 depends on k -parameters $c_1, \dots, c_k$

IV [optionally]
 if $a_0, a_1, \dots, a_{k-1}$ are given
 parameters $c_1, c_2, \dots, c_k$ are computed
 from k -equations in k -unknowns

Example: $IH \begin{cases} a_n - 2a_{n-1} = 3^n * \\ a_0 = 4 (**), (k=1) \end{cases}$

I) $r - 2 = 0$ (characteristic eqn
 for $h_n - 2h_{n-1} = 0$)

$r = 2$ $\boxed{h_n = C \cdot 2^n}$

II) Prediction method: since $f(n) = 3^n$
 we guess the solution (see table
 at the end of this lecture) as:

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$b_n = A 3^n$ $A = ?$ ATTENTION $3 \neq r = 2$
 as $f(n) = r^n \Rightarrow b = A r^n$ if $r \neq r$ we treat as multiplicity 2
 so we substitute into inhomogeneous

equation(*) $A 3^n - 2 A 3^{n-1} = 3^n$

$$3A - 2A = 3$$

$$\boxed{A = 3}$$

$$\Rightarrow \boxed{b_n = 3^{n+1}}$$

III $a_n = h_n + b_n = \boxed{C 2^n + 3^{n+1} = a_n}$
 general form of the solution to (*) depending on 1-parameter ($k=1$) here

IV Since we know $a_0 = 4$

$$4 = a_0 = C 2^0 + 3^{0+1}$$

$$\Downarrow$$

$$C = 1$$

$$\Downarrow$$

a unique solution to (*) & (**)

$$\boxed{a_n = 3^{n+1} + 2^n} \Rightarrow \boxed{a_n = O(3^n)}$$

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Example Hanoi problem:

$$\begin{cases} T_n = 2T_{n-1} + 1 & (*) \\ T_1 = 1 & (**) \end{cases}$$

I) $h_n - 2h_{n-1} = 0$
 $r - 2 = 0$ (characteristic polynomial.)
 $\underline{r=2} \Rightarrow \underline{h_n = C \cdot 2^n}$

II) as $f(n) = 1 = 1^n = \bar{r}^n$
 prediction method suggests to
 search for $b_n = A \cdot \bar{r}^n = A$
 (as $\bar{r} \neq 2$)
 To compute A we plug A into (*)

$$A = 2A + 1 \Rightarrow A = -1$$

$$\Rightarrow \underline{b_n = -1}$$

III) $a_n = C 2^n - 1$ general solution to (*) depending on 1 parameter C ($k=1$)

IV) By (**) $1 = a_1 = C 2^1 - 1 \Rightarrow \underline{C=1}$
 $\underline{a_n = 2^n - 1}$
 $\underline{a_3 = 7}$ we had that. See lecture 5.

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Example: (*) $a_n + a_{n-1} - 6a_{n-2} = 2^n - 1$

$\uparrow$
 $f(n)$

I] $h_n + h_{n-1} - 6h_{n-2} = 0$

$\Downarrow$

$$r^2 + r - 6 = 0$$

$$r_1 = -3 \quad \& \quad r_2 = 2$$

$$\Rightarrow \boxed{h_n = C_1(-3)^n + C_2 2^n}$$

II Prediction method: $2^n - 1^n = f(n)$

As $\begin{matrix} \bar{r} \\ \parallel \\ 2 \end{matrix} = \begin{matrix} r_2 \\ \parallel \\ 2 \end{matrix}$ prediction method

$$\text{so } \boxed{b_n = Cn 2^n + D \cdot 1^n}$$

We plug b_n into (*) to compute C & D

$$(Cn 2^n + D) + (C(n-1)2^{n-1} + D) - 6(C(n-2)2^{n-2} + D) = 2^n - 1$$

$$\Downarrow \quad \boxed{\frac{5}{2} C 2^n - 4D = 2^n - 1} \quad \forall n \in \mathbb{N}$$

$$\frac{5}{2}C = 1 \quad \& \quad -4D = -1$$

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$$\Rightarrow C = \frac{2}{5} \quad \& \quad D = \frac{1}{4}$$

special solution to $b_n = \frac{2}{5}n2^n + \frac{1}{4}$
inhomogeneous eqn (*)

III $a_n = h_n + b_n = C_1(-3)^n + C_22^n + \frac{2}{5}n2^n + \frac{1}{4}$

as we do not have $\hat{a}_0$ & $\hat{a}_1$
this is a final answer $(a_n = \Theta(3^n))$

We finish this lecture by hinting how to guess b_n given special forms of $f(n)$ in:

$$a_n = \sum_{i=1}^K c_i a_{n-i} + f(n)$$

Note that in f we may have $\bar{r}^n$ with $\bar{r} = r$ (one of the roots of characteristic polynomials)

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$f(n)$		prediction of br special solution
c	$\Rightarrow$	A constant
n	$\Rightarrow$	$A_1 n + A_0$
n^2	$\Rightarrow$	$A_2 n^2 + A_1 n + A_0$
$n^t \quad t \in \mathbb{Z}_+$	$\Rightarrow$	$A_t n^t + A_{t-1} n^{t-1} + \dots + A_1 n + A_0$
$r^n \quad r \in \mathbb{R}$	$\Rightarrow$	$A r^n$
$\sin(\alpha n)$	$\Rightarrow$	$A \sin(n\alpha) + B \cos(n\alpha)$
$\cos(\alpha n)$	$\Rightarrow$	$A \sin(n\alpha) + B \cos(n\alpha)$
$n^t r^n$	$\Rightarrow$	$r^n (A_t n^t + \dots + A_1 n + A_0)$
$r^n \sin(\alpha n)$	$\Rightarrow$	$A r^n \sin(n\alpha) + B r^n \cos(n\alpha)$
$r^n \cos(\alpha n)$	$\Rightarrow$	$A r^n \sin(n\alpha) + B r^n \cos(n\alpha)$

Things gets trickier if the summand $f_1(n)$ of $f(n)$ is a constant multiple of a solution of the associated homogeneous system. This happens e.g. when $f(n)$ contains summand as $c r^n$ or $(c_1 + c_2 n) r^n$ & r is a root of characteristic polynomial

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Then we multiply such solution by the smallest power of n say n^s for which no summand $n^s f_1(n)$ is a solution to the associated homogeneous relation

Method based on I - IV

has its:

advantages:

- the algorithmic set of steps to transform implicit form into explicit form of the recursive formula permitting e.g. to find $a_n = O(f(n))$

disadvantages:

- applies only to the schemes
$$a_n = \sum_{i=1}^k c_i a_{n-i} + f(n)$$
- relies on finding roots of characteristic polynomial (can be hard)